# Harmonic Balance Analysis of Nonlinear Dielectrics Filled Waveguide Devices

Laurent Ntibarikure?, *Student Member, IEEE,* Giuseppe Pelosi, *Fellow, IEEE,*
Lorenzo Rossi?, and Stefano Selleri, *Senior Member, IEEE*

***Abstract*—This letter presents the computation of high order harmonic components of the time-harmonic field exciting waveguide structures partially filled with nonlinear dielectrics. The approach presented employs the Harmonic Balance method to perform a multi-harmonic Finite Element analysis, and the solution of the nonlinear system is handled either with a Picard or a relaxed iteration, depending on the degree of the nonlinear behavior. Since the nonlinear loop is particularly time demanding, a domain decomposition method is here proposed to speed-up the system solution, restricting the iteration loop to a small subdomain.**

***Index Terms*—Harmonic Balance, Finite Elements, Domain Decomposition, nonlinear dielectrics, waveguide filter**

## I. Introduction

TODAY's microwave and millimeter wave passive devices design require, when high power densities are involved, the modeling of material nonlinear properties to accurately predict the steady state electromagnetic field. Among the available techniques, a time-stepping approach to the solution of Maxwell's equations in such cases results to be computationally prohibitive [1]. An alternative time-harmonic approach, based on the finite element (FE) solution of a partial differential equation (PDE) of time-harmonic variable, provides the field quantities at a given frequency. Unfortunately, a single frequency is not sufficient to include all relevant phenomena, high order harmonics being generated from the interaction of the electromagnetic field with the nonlinear materials. Incorporating a larger set of harmonics in the FE formulation by using the Harmonic Balance Finite Element (HBFE) method [2], [3], also known as Multi-Harmonic Finite Element method [4]–[6], provides a fast and accurate solution to the problem as long as the harmonics required to affordably represent the field is not excessive. The nonlinearity of the PDE is tackled by an iterative solution of a spatially and spectrally coupled large-scale linear system updated at each iteration with a new linearization such that the computed solutions converge, within a prescribed tolerance, to the sought solution. An algebraic Domain Decomposition technique as proposed in [7] improves computational timings, especially when the region occupied by nonlinear materials is rather small.

In this letter, we focus on the analysis of waveguide devices filled with dielectrics that present nonlinear scalar relative permittivity. In section II, we present the Harmonic Balance Finite Element method formulation for the solution of the vector Helmholtz equation for the electric field, within a domain comprising nonlinear dielectrics. In section III, simulation results of a broadband E-plane bandpass filter filled with Kerr-type material are shown, and the performances of a domain decomposition algorithm used to enhance the computations are discussed.

Fig. 1. Sketch of the domain $\Omega = \Omega_1 \cup \Omega_2$.

## II. Formulation

The formulation given in this section holds for the case of a generic electromagnetic problem defined on a closed domain as depicted in Fig. 1. The domain $\Omega$, with boundary $\Gamma_0$, can be either defined on $\mathbb{R}^2$ or $\mathbb{R}^3$. For the sake of simplicity, the examples presented in the next section are analyzed in $\mathbb{R}^2$, allowing for straightforward implementation. $\Omega$ is divided into a large part, $\Omega_1$, with linear scalar relative permittivity $\bar{\varepsilon}_r(\mathbf{r})$ and a small part, $\Omega_2$, with a nonlinear relative permittivity $\tilde{\varepsilon}_r(\mathbf{E}(\mathbf{r}))$. The scalar permittivity in $\Omega$ can be expressed as

$$\varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) = \begin{cases} \bar{\varepsilon}_r(\mathbf{r}), & \mathbf{r}\in\Omega_1 \\ \tilde{\varepsilon}_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) = \bar{\varepsilon}_r(\mathbf{r}) + \mathcal{L}(\mathbf{E}(\mathbf{r})), & \mathbf{r}\in\Omega_2 \end{cases} \tag{1}$$

where $\mathbf{E}(\mathbf{r})$ is the electric field in $\mathbf{r}$ and $\mathcal{L}(\cdot)$ denotes a generic scalar operator that describes the analytically known nonlinear behavior. The electric field satisfies, within $\Omega$, the vector Helmholtz equation

$$\nabla\times\left(\frac{1}{\mu_r}\nabla\times\mathbf{E}(\mathbf{r})\right) - {k_0}^2\varepsilon_r\left(\mathbf{E}(\mathbf{r}),\mathbf{r}\right)\mathbf{E}(\mathbf{r}) = 0 \tag{2}$$

with suitable boundary conditions on $\Gamma_0$. For the example shown in section III, $\Gamma_0$ is considered partly perfect electric conductor, partly a modal expansion of the field at the wave ports as developed in [8].

We proceed with the standard FE Galerkin framework: the domain $\Omega$ is discretized into a set of non overlapping finite elements and the electric field is approximated as a linear combination of vector basis functions $\boldsymbol{\alpha}_n$

$$\mathbf{E}(\mathbf{r}) = \sum_{n=1}^{N} E_n\boldsymbol{\alpha}_n(\mathbf{r}) \tag{3}$$

Then by testing (2) over $\Omega$ with the weighting functions $\mathbf{W}_m$, chosen the same as the shape functions $\boldsymbol{\alpha}_n$, we obtain the

L. Ntibarikure?, G. Pelosi and S. Selleri are with the Department of Electronics and Telecommunications, University of Florence, I-50139 Florence, Italy (e-mail: stefano.selleri@unifi.it).

L. Rossi?

following weak form:

$$\int_{\Omega} \nabla \times \mathbf{W}_m(\mathbf{r}) \cdot \frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) \; d\Omega - k_0^2 \int_{\Omega} \mathbf{W}_m(\mathbf{r}) \cdot \varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) \mathbf{E}(\mathbf{r}) \; d\Omega + \oint_{\Gamma_0} \mathbf{W}_m(\mathbf{r}) \cdot \hat{n} \times \frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) \; d\Gamma = 0 \tag{4}$$

We introduce the multi-harmonic dependence of the HBFE method, by approximating the electric field as a truncated Fourier series [5]

$$\hat{\mathbf{E}}(\mathbf{r},t) = \sum_{p=1}^{P} \mathbf{E}_p^{(s)}(\mathbf{r}) \sin(p\omega t) + \mathbf{E}_p^{(c)}(\mathbf{r}) \cos(p\omega t) \tag{5}$$

where $\omega$ is the fundamental angular frequency. The Fourier expansion is devoid of a dc-term as a static field does not represent a solution to (2) for $k_0 > 0$. $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$ are the harmonic complex amplitudes that expand the finite element coefficients $E_n$, and they must individually satisfy (2).

The scalar relative permittivity can also be approximated as a truncated Fourier series

$$\hat{\varepsilon}_r(\mathbf{E}(\mathbf{r}),\mathbf{r},t) = \varepsilon_{r_0} + \sum_{k=1}^{K} \varepsilon_{r_k}^{(s)} \sin(k\omega t) + \varepsilon_{r_k}^{(c)} \cos(k\omega t) \tag{6}$$

and the coefficients of the expansion are given by

$$\varepsilon_{r_0} = \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) \; dt \tag{7}$$

$$\varepsilon_{r_k}^{(s)} = \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) \sin(k\omega t) \; dt \tag{8}$$

$$\varepsilon_{r_k}^{(c)} = \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) \cos(k\omega t) \; dt \tag{9}$$

with $k = 1 \ldots K$. The approximation orders $P$ and $K$ of, respectively, the electric field and the scalar permittivity, depend on the specific device analyzed and on the intensity of the incident field. These can be set by choosing an appropriate convergence criterion of the power associated to the fundamental as the number of harmonic coefficients is increased [9].

We proceed substituting $\mathbf{E}(\mathbf{r})$ and $\varepsilon_r(\mathbf{r})$ by their respective harmonic approximations $\hat{\mathbf{E}}(\mathbf{r},t)$ and $\hat{\varepsilon}_r(\mathbf{r},t)$ in (4), then testing the equation with $\sin(m\omega t)$ and $\cos(m\omega t)$, $m = 1 \ldots M$, and integrating over the time period of the fundamental $\left[0, \frac{2\pi}{\omega}\right]$ gives a large-scale linear system of the form:

$$\begin{bmatrix} a_{11}^{(1s1s)} & a_{11}^{(1s1c)} & \ldots & a_{1N}^{(1sPc)} \\ a_{11}^{(1c1s)} & a_{11}^{(1c1c)} & \ldots & a_{1N}^{(1cPc)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1}^{(Pc1s)} & a_{N1}^{(Pc1c)} & \ldots & a_{NN}^{(PcPc)} \end{bmatrix} \begin{bmatrix} E_1^{(1s)} \\ E_1^{(1c)} \\ \vdots \\ E_N^{(Pc)} \end{bmatrix} = \begin{bmatrix} b_1^{(1s)} \\ b_1^{(1c)} \\ \vdots \\ b_N^{(Pc)} \end{bmatrix} \tag{10}$$

The product that occurs in the second integral of (4), between the approximated field and the permittivity, induces coupling between the harmonic components. However, as the PDE we seek to solve is free of first order time derivative, there is no coupling between $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$. Thus, we may retain either the sine or the cosine harmonic function to excite the electromagnetic device and to express the multi-harmonic behavior of the field, halving in such a way the system dimensions. The order of the resulting HBFE system is $NP$. In practical analyses, this number can become considerable, and as introduced previously, the solutions have to be computed iteratively, building at each step a new linearized system, until convergence of the solution is achieved.

The chosen nonlinear solution is based on a relaxed iteration, with $\gamma \in [0, 1]$ the relaxation coefficient, and $\gamma = 1$ constitutes the particular case of the Picard iteration. The permittivity is assumed initially to be the one in the absence of electric field:

$$[A_{([E]^0)}][E]^1 = [b_{([E]^0)}] \tag{11}$$

where $[E]^0$ is a null vector of unknowns. The solution of the first linear system of (11) provides expansion coefficients that will be used to compute a new value of the permittivity:

$$[A_{(\gamma[E]^{q-1}+(1-\gamma)[E]^{q-2})}][E]^q = [b_{(\gamma[E]^{q-1}+(1-\gamma)[E]^{q-2})}] \tag{12}$$

with $q \geq 2$. The process is repeated until the relative error between the updated solution and the previous one, in the sense of the euclidean norm, is less than a prescribed value $\tau$:

$$\frac{\|[E]^q - [E]^{q-1}\|}{\|[E]^q\|} \leq \tau \tag{13}$$

For the example of section III, $\tau = 10^{-6}$ have been chosen. Highly nonlinear materials and high intensity impressed fields may hamper the convergence of the nonlinear iteration, due to the oscillation of the solutions around the expected values. Lowering the value of $\gamma$ has the effect to damp that oscillation, hence rendering the iteration more robust but inevitably reducing the convergence rate [10].

To enhance the computational efficiency, a substructuring domain decomposition approach is used [7], [11]–[13]. The multi-harmonic unknowns vector $[E]$ defined by the HBFE method over $\Omega$ can be split into two vectors $[E^{(1)}]$ and $[E^{(2)}]$ containing the unknowns belonging to interior points in $\Omega_1$ and $\Omega_2$, respectively. Unknowns belonging to the boundaries $\Gamma$ are placed in a third vector $[E^c]$. The HBFE system can then be recast in the form:

$$\begin{bmatrix} [M^{(1)}] & 0 & [P^{(1)}] \\ 0 & [M^{(2)}_{([E]^{q-1})}] & [P^{(2)}_{([E]^{q-1})}] \\ [Q^{(1)}] & [Q^{(2)}_{([E]^{q-1})}] & [C_{([E]^{q-1})}] \end{bmatrix} \begin{bmatrix} [E^{(1)}]^q \\ [E^{(2)}]^q \\ [E^{(c)}]^q \end{bmatrix} = \begin{bmatrix} [b^{(1)}] \\ [b^{(2)}_{([E]^{q-1})}] \\ [b^{(c)}_{([E]^{q-1})}] \end{bmatrix} \tag{14}$$

where $[M^{(1)}]$ contains the HBFE coefficients of the linear system related to the unknowns $[E^{(1)}]$ fully internal to $\Omega_1$, hence with null harmonic coupling coefficients. $[M^{(2)}]$ contains coefficients for the nonlinear sub-domain $\Omega_2$. $[P^{(\cdot)}]$ and $[Q^{(\cdot)}]$ represent couplings between the interior unknowns and the ones on the boundaries which are collected in $[E^{(c)}]$. By using the Schur complement $[S_{([E]^{q-1})}]$, constructed with a Gauss elimination-type strategy, the boundary unknowns $[E^{(c)}]^q$ of (14) are computed by solving the reduced system

$$[S_{([E]^{q-1})}][E^{(c)}]^q = [d]^{q-1} \tag{15}$$

Fig. 2. Cross section of a broadband E-plane bandpass filter realized in an empty WR6 rectangular waveguide [14]. Measures are given in $\mu m$.

Fig. 3. Behavior of the $\mathrm{TE}_{10}$ spectral response as the number of harmonics retained for the HBFE increases ($\mathrm{E_i} = 10\mathrm{kV/m}$).

Fig. 4. $|\mathrm{E}_y|$ distribution within the filter at fundamental ($f_0 = 146$ GHz), $3^{\mathrm{rd}}$ and $5^{\mathrm{th}}$ order harmonics ($\mathrm{E_i} = 10\mathrm{kV/m}$).

Fig. 5. $S_{21}$ amplitude responses of the nonlinear filter given for several values of $\mathrm{E_i}$ compared to the linear case. The field is approximated up to the $5^{\mathrm{th}}$ harmonic.

Fig. 6. Acceleration spectrum obtained for a HBFE systems of several harmonic orders. The computations are made for $\mathrm{E_i} = 10\mathrm{kV/m}$.

where

$$\begin{aligned}[S_{([E]^{q-1})}] = [C_{([E]^{q-1})}] - [Q^{(1)}][M^{(1)}]^{-1}[P^{(1)}] \\ -[Q^{(2)}_{([E]^{q-1})}][M^{(2)}_{([E]^{q-1})}]^{-1}[P^{(2)}_{([E]^{q-1})}]\end{aligned} \quad (16)$$

and

$$\begin{aligned}[d]^k = [b^{(c)}_{([E]^{q-1})}] - [Q^{(1)}][M^{(1)}]^{-1}[b^{(1)}] \\ -[Q^{(2)}_{([E]^{q-1})}][M^{(2)}_{([E]^{q-1})}]^{-1}[b^{(2)}_{([E]^{q-1})}]\end{aligned} \quad (17)$$

The difference operators in (16) and (17) are to be intended as an iterative assembling of local matrices in a global one [12]. Once $[E^{(c)}]^q$ is known, local solutions are given by

$$[E^{(1)}]^q = [M^{(1)}]^{-1}\left([b^{(1)}] - [P^{(1)}][E^{(c)}]^q\right) \quad (18)$$

$$\begin{aligned}[E^{(2)}]^q = [M^{(2)}_{([E]^{q-1})}]^{-1}\Big([b^{(2)}_{([E]^{q-1})}] \\ -[P^{(2)}_{([E]^{q-1})}][E^{(c)}]^q\Big)\end{aligned} \quad (19)$$

In order to solve the HBFE system and compute the steady state field, every single submatrix is assembled at first sight, then, within the iteration loop, only the submatrices related to $\Omega_2$ and its boundary are to be updated. The efficiency of this DD technique relies on the fragmentation of the matrices and the subsequent computation of partial solutions. When the number of coefficients related to the nonlinear media is small, noticeable improvements can be achieved.

## III. Numerical Results

In this section, the analysis of a millimeter wave broadband filter [14] is performed and the computational results are shown. The filter is realized within a standard WR6 waveguide by properly designed metalizations on a 6040.5 $\mu m$ long dielectric slab, mounted along the E-plane at the middle of the width range. A sketch of the H-plane cross section of the filter, plane of the performed 2D HBFE analysis, is shown in Fig. 2. First order nodal elements are used and the domain discretization is such that 2470 degrees of freedom are obtained, 42 belonging to the nonlinear subdomain $\Omega_2$, 456 to the boundaries and the remaining 1972 to the linear subdomain $\Omega_1$. The metalizations are considered perfectly conducting, thus imposing the appropriate Dirichelet conditions. The continuity of the field at the ports is imposed through a modal expansion which, for the geometry of the problem, consists of $\mathrm{TE}_{n0}$ modes, $n \in \mathbb{Z}$, where $n \leq 10$ have been chosen. The incident field at Port 1 is only of a $\mathrm{TE}_{10}$ mode and $\mathrm{E_i}$ is the associated electric field maximum amplitude.

The dielectric, enclosed in $\Omega_2$, presents a Kerr-type nonlinearity such that [15]

$$\tilde{\varepsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \bar{\varepsilon}_r(\mathbf{r}) + \alpha_2\, |\mathbf{E}(\mathbf{r})|^2, \qquad \mathbf{r} \in \Omega_2 \quad (20)$$

where $\bar{\varepsilon}_r(\mathbf{r}) = 2.1$ and $\alpha_2$ is chosen of $1.625\ 10^{-10}\ \mathrm{m^2/V^2}$ according to [16]. The Kerr-like behavior of the permittivity induces the generation of odd order harmonics, thus the even ones have been neglected in the truncated Fourier expansion. Furthermore, as we choose a sinusoidal excitation, only $\sin(m\omega t)$ related coefficients of the electric field are retained.

We first analyze the effect of the number of harmonics retained on the filter's spectral response as shown in Fig. 3. For the nonlinearity degree represented by $\mathrm{E_i}$ =10 kV/m and the chosen $\alpha_2$, the Picard iteration was sufficient to tackle the iteration loop. As we can see, the inclusion of the $3^{\mathrm{rd}}$ harmonic is crucial for proper bandwidth evaluation of the device. The field distribution within the device at 146 GHz, $3^{\mathrm{rd}}$ and $5^{\mathrm{th}}$ harmonics are shown in Fig. 4.

To see whether the nonlinear dielectrics influences the spectral response of the device, the intensity of the field has been set to 5 kV/m, 10 kV/m and 15 kV/m corresponding to 18 mW, 72 mW and 162 mW of incident power, respectively, which are common power values encountered in D-band applications [17]. The results, compared to those of a linear device ($\alpha_2 = 0$), are shown in Fig. 5. The field is approximated up to the $5^{\mathrm{th}}$ harmonic. There is an evident shift the bandpass range to lower frequencies as higher power densities are involved. $\mathrm{E_i}$ =15 kV/m has required a relaxed iteration with $\gamma = 0.1$. In this case, the abrupt variation of $|S_{21}|$ around 139.5 GHz points out that the harmonic order used is not sufficient to accurately analyze the device. Furthermore, a finer mesh would be necessary to validate the inclusion of higher order harmonics.

The efficiency of the DD technique versus the full domain solution is analyzed introducing the *acceleration*

$$a \backsimeq \frac{I \cdot (T_\Omega)}{T_{\Omega_1} + I \cdot (T_{\Omega_2} + T_\Gamma)} \quad (21)$$

where $I$ is the number of iterations required by the nonlinear loop and $T_\Omega$, $T_{\Omega_1}$, $T_{\Omega_2}$ and $T_\Gamma$ are assembling and solving times. Fig. 6 depicts the spectral acceleration obtained increasing the number of harmonics for $\mathrm{E_i} = 10\mathrm{kV/m}$. In this case, as stated before, at least a $3^{\mathrm{rd}}$ harmonic is required for accurate solution. In the average, higher order HBFE systems have lower acceleration, the matrices dimensions related to nonlinear media increasing. With a $3^{\mathrm{rd}}$ order system, the acceleration varies in the range [2.51, 9.15] such that the overall acceleration is of 4.18. In this case, the DD HBFE solution uses 24% of the time required by the full domain HBFE solution.

## IV. Conclusion

An Harmonic Balance approach to the FE solution of nonlinear dielectrics filled waveguide devices has been presented and the accuracy of the multi-harmonic solution discussed.A domain decomposition technique has been used to speed-up the computations and the efficiency of the algorithm has been evaluated.

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